13 Observability and monitoring

This chapter covers

- Logging with Spring Boot, Loki, and Fluent Bit
- Using health probes with Spring Boot Actuator and Kubernetes
- Producing metrics with Spring Boot Actuator, Prometheus, and Grafana
- Configuring distributed tracing with OpenTelemetry and Tempo
- Managing applications with Spring Boot Actuator

In the previous chapters, you learned about several patterns and technologies you can use to build secure, scalable, and resilient applications. However, we still lack visibility into the Polar Bookshop system, especially when something goes wrong. Before going to production, we should ensure our applications are observable and that the deployment platform provides all the tools needed to monitor and gain insights into the system.

Monitoring involves checking the telemetry available for the application and defining alerts for known failure states. *Observability* goes beyond that and aims at reaching a state where we can ask arbitrary questions about the system without knowing the question in advance. The product team should ensure their applications expose relevant information; and the platform team should provide an infrastructure for consuming that information and asking questions about their operations.

As you'll remember from chapter 1, *observability* is one of the properties of cloud native applications. Observability is a measure of how well we can infer the internal state of an application from its outputs. In chapter 2, you